

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Machine Learning
ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO2 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	AIML	Date:	6.8.2026

Code of Conduct

I P. DEEKSHANA / 192425104 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

QUESTIONS: 5

Total Marks: 50

Annexure A

Scenario-Based Assignment

Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question: Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question: Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex real-world datasets.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question: Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question: Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question: Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

1. Machine Learning vs Deep Learning

A healthcare company aims to detect pneumonia from chest X-ray images. Initially, it considers Machine Learning (ML) with manually extracted image features, but after collecting millions of X-ray images, it switches to Deep Learning (DL).

Aspect	Machine Learning	Deep Learning
Data Requirement	Performs well with small to medium-sized datasets.	Requires very large datasets to achieve high accuracy.
Feature Extraction	Features such as edges, texture, and shape must be manually designed by experts.	Automatically learns relevant features directly from raw X-ray images.
Computational Resources	Can be trained on standard CPUs with lower computational cost.	Requires high-performance GPUs/TPUs due to complex neural networks.
Accuracy	Performance depends heavily on manually selected features.	Generally provides higher accuracy by learning complex image patterns.
Scalability	Difficult to improve performance as data grows.	Performance usually improves with larger datasets and deeper models.
Real-World Challenges	Requires domain expertise for feature engineering and may miss subtle disease patterns.	Needs large labeled datasets, long training time, high computing cost, and model interpretability remains challenging.

Justification: Deep Learning is a better choice because:

- Millions of X-ray images provide sufficient data for training deep neural networks.
- It automatically learns important medical features without manual intervention.
- Convolutional Neural Networks (CNNs) can detect subtle pneumonia patterns that traditional ML may overlook.
- DL models achieve higher diagnostic accuracy and better generalization.
- Despite requiring expensive hardware and longer training time, the improved performance makes DL suitable for real-world medical diagnosis.

Conclusion: Deep Learning is preferred for pneumonia detection because it automatically extracts complex image features, handles large datasets efficiently, and provides higher diagnostic accuracy than traditional Machine Learning.

2. Representation Learning

Representation Learning is a Deep Learning technique in which a model automatically learns meaningful features from raw data instead of relying on manually designed features.

Working:-

For the retail company:

- Raw customer transaction records are directly provided to the neural network.
- Hidden layers learn increasingly complex buying patterns.
- The learned representations capture customer preferences, purchasing habits, and product relationships automatically.

Traditional Feature Engineering vs Representation Learning

Traditional Feature Engineering	Representation Learning
Features are manually created by experts.	Features are learned automatically from data.
Time-consuming and domain-dependent.	Minimal manual effort required.
Limited ability to capture complex relationships.	Learns highly complex and non-linear relationships.
Performance depends on feature quality.	Performance improves with more data.

Advantages

- Eliminates manual feature engineering.
- Discovers hidden customer behavior patterns.
- Handles very large and complex datasets.
- Learns hierarchical feature representations.
- Improves prediction accuracy.
- Easily adapts to changing customer behavior.

Suitable Techniques

- Autoencoders
- Restricted Boltzmann Machines (RBMs)
- Deep Belief Networks (DBNs)
- Self-Supervised Learning

- Contrastive Learning

Conclusion: Representation Learning enables the retail company to automatically discover meaningful customer purchasing patterns from raw transactional data, reducing manual effort while improving scalability and prediction performance.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system.

Approach 1: Wide Network

- Single hidden layer
- Thousands of neurons

Advantages

- Simpler architecture.
- Faster forward propagation.
- Can approximate many functions.

Disadvantages

- Large number of parameters.
- High memory usage.
- More prone to overfitting.
- Learns limited hierarchical features.

Approach 2: Deep Network

- Multiple hidden layers
- Fewer neurons per layer

Advantages

- Learns hierarchical features.
- Better feature abstraction.
- Requires fewer parameters for similar complexity.
- Better generalization.
- Higher recognition accuracy.

Disadvantages

- Longer training time.
- Requires more computational resources.

- More difficult to tune.

Comparison

Aspect	Wide Network	Deep Network
Hidden Layers	One	Multiple
Feature Learning	Limited	Hierarchical
Parameters	Very high	Moderate
Memory Usage	High	Moderate
Computational Complexity	Lower depth but many neurons	Higher depth but efficient representation
Generalization	More overfitting	Better generalization
Performance	Moderate	Higher accuracy

Recommendation: A deep neural network is recommended because:

- Learns simple features (edges) in early layers.
- Learns shapes and digits in deeper layers.
- Provides better feature extraction.
- Generalizes well to unseen handwritten digits.

Conclusion: Deep architectures are more suitable for handwritten digit recognition because hierarchical learning captures increasingly complex visual features while reducing overfitting.

4. Activation Functions

Different activation functions solve different training problems.

Activation Function	Characteristics	Suitable Layer
Sigmoid	Output between 0 and 1; suffers from vanishing gradient.	Binary output layer
Tanh	Output between -1 and 1; zero-centered but still has vanishing gradient.	Hidden layers (limited use)
ReLU	Fast computation; reduces vanishing gradient; may produce dead neurons.	Hidden layers
Leaky ReLU	Small negative slope prevents dead neurons.	Hidden layers
ELU	Negative outputs improve learning stability and faster convergence.	Hidden layers
Softmax	Produces probability distribution across classes.	Multi-class output layer

Layer-wise Recommendation

Input Layer

- No activation function.

Hidden Layers

- Leaky ReLU or ELU
- Prevent dead neurons.
- Faster convergence.
- Better gradient flow.

Output Layer

- Sigmoid → Binary classification.
- Softmax → Multi-class classification.

Problem-wise Solution

Problem	Solution
Slow Convergence	ReLU, ELU
Vanishing Gradient	ReLU, Leaky ReLU, ELU
Dead Neurons	Leaky ReLU, ELU
Binary Classification	Sigmoid
Multi-class Classification	Softmax

Conclusion: For image classification, Leaky ReLU or ELU should be used in hidden layers, while Softmax is ideal for the final multi-class output layer because it produces normalized class probabilities.

5. Unsupervised Training of Neural Networks

Unsupervised learning trains neural networks using unlabeled data, allowing the model to automatically discover hidden structures and meaningful representations.

Learning Process

1. Collect large unlabeled image datasets.
2. Feed raw images into the neural network.
3. The model identifies similarities and patterns.
4. Hidden layers learn useful feature representations.
5. Learned features can later be used for classification or other downstream tasks.

Advantages over Supervised Learning

Supervised Learning	Unsupervised Learning
Requires labeled data.	Uses unlabeled data.
Labeling is expensive.	No manual labeling required.
Limited by available labels.	Can utilize massive datasets.
Higher annotation cost.	Lower overall cost.

Suitable Techniques

- **Autoencoders** – Learn compressed feature representations by reconstructing input data.
- **Variational Autoencoders (VAEs)** – Learn meaningful latent representations useful for generation and feature learning.
- **Restricted Boltzmann Machines (RBMs)** – Discover hidden structures in data.
- **Deep Belief Networks (DBNs)** – Stack RBMs to learn hierarchical features.
- **Self-Supervised Learning** – Creates learning tasks from the data itself to learn rich representations.
- **Contrastive Learning** – Learns representations by bringing similar images closer and pushing dissimilar ones apart.

Benefits

- Utilizes billions of unlabeled images efficiently.
- Reduces annotation cost and time.
- Learns robust feature representations.
- Improves performance on downstream tasks after fine-tuning.
- Scales well to continuously growing datasets.

Conclusion: Unsupervised neural network training is ideal for the social media platform because it can learn meaningful visual representations from vast amounts of unlabeled images, significantly reducing labeling costs while providing high-quality features for future tasks such as image classification, recommendation, and content retrieval.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____